

Self-focusing and Ponderomotive Forces of Coherent Electric Waves: A Mechanism for Cytoskeleton Formation and Dynamics

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Coherent electric waves have been predicted to exist in biological systems.⁽¹⁾ We try to link the consequences of their propagation in the cell medium to some peculiar features of cell architecture and organization such as cytoskeleton formation and dynamics.⁽²⁾ This will give a consistency argument in favour of the existence of coherent electric waves in living systems.⁽³⁾

The general ponderomotive force produced by an electric field $\vec{E}$ and a magnetic field $\vec{H}$ acting on a dielectric with polarization $\vec{P}$ and magnetization $\vec{M}$ is given by⁽⁴⁾

$$\vec{f} = \rho \vec{E} + \frac{1}{c} \vec{j} \times \vec{H} + (\vec{P} \cdot \nabla) \vec{E} + (\vec{M} \cdot \nabla) \vec{H} + \frac{1}{c} \frac{\partial \vec{P}}{\partial t} \times \vec{H} - \frac{1}{c} \frac{\partial \vec{M}}{\partial t} \times \vec{E} \quad (1)$$

In a first approach we will drop the terms depending on the magnetization $\vec{M}$ (which is expected to be important only in a specialized class of cells) and we will also omit the linear term arising from the "true" current source $\vec{j}$. Therefore

$$\vec{f} = \rho \vec{E} + (\vec{P} \cdot \nabla) \vec{E} + \frac{1}{c} \frac{\partial \vec{P}}{\partial t} \times \vec{H} \quad (1')$$

In eq. (1') we have the usual linear term $\rho \vec{E}$, the nonlinear term, called the "gradient force" $(\vec{P} \cdot \nabla) \vec{E}$, and the term of "radiation pressure" $\frac{1}{c} \frac{\partial \vec{P}}{\partial t} \times \vec{H}$.

Since coherent waves can bring very large values of $\vec{E}$ (of the order of 10^7 V/m), the nonlinear terms of eq. (1') are expected to dominate on the linear one.

Consequently, we will drop the first term in the right hand side of eq. (1').

The gradient term can be written as⁽⁵⁾

$$(\vec{p} \cdot \nabla) \vec{E} = - \text{grad } p_0(\rho, T) + \frac{1}{8\pi} \text{grad } E^2 \rho \left(\frac{\partial \epsilon}{\partial \rho} \right)_T +$$

$$- \frac{1}{8\pi} E^2 \text{grad } \epsilon$$
(2)

where ϵ is the dielectric permittivity of the medium and $p_0(\rho, T)$ is the pressure in the medium in the absence of an external field, at the same volume and temperature T . The pressure gradient term, although relevant in many applications, can be disregarded in a microscopic approach. Moreover, we will use an approximation valid for gaseous media and dilute solutions (as the cytoplasm can be considered in a rough approximation):

$$\rho \frac{\partial \epsilon}{\partial \rho} \sim (\epsilon - 1)$$
(3)

After trivial manipulations one gets from eq. (1')

$$\vec{f} = \frac{1}{8\pi} (\epsilon - 1) \nabla E^2 + \frac{1}{c} \frac{\partial \vec{p}}{\partial t} \times \vec{H}$$
(4)

The first term in the right hand side of eq. (4) shows that a high power, strong directional electric wave

$$\vec{E}(\vec{r}, t) = \frac{1}{2} \vec{A}(\vec{r}, t) \exp [i(\omega t - \vec{k} \cdot \vec{r})] + \text{c.c.}$$

can produce condensation or rarefaction in a medium according to the sign of the coefficient of ∇E^2 . This coefficient is a function both of the frequency ω of the beam and of the typical resonance frequencies ω_{ok} of the dielectric. It has been shown⁽⁶⁾ that

$$\frac{1}{8\pi} (\epsilon - 1) \nabla E^2 = C \sum_k \left[\frac{\omega_{ok}^2 - \omega^2}{(\omega_{ok}^2 - \omega^2)^2 + \Gamma_k^2} \right] \nabla E^2$$
(5)

where C is some positive constant and Γ_k is the damping of the k -th oscillator. This gradient force is important when the incoming frequency ω approaches ω_o , one of the frequencies of the medium. An attractive (repulsive) force in the direction transverse to the propagation axis of the beam will then be present when $\omega_o > \omega$ ($\omega_o < \omega$).

The gradient force produces then a waveguide in the medium along the intense and directional electric beam, where a condensation of matter takes place. This process eliminates the geometric and diffraction divergences of the beam, leading to a very efficient propagation within the waveguide ("self-focusing").

In order to discuss this kind of propagation, the gradient force can be expressed in terms of the properties of the refractive index of the nonlinear medium⁽⁷⁾. On the basis of the Kerr properties of protein solutions⁽⁸⁾, we will consider a "cubic"

medium, i.e. $n = n_0 + n_2 |A|^2$ ($\epsilon = \epsilon_0 + \epsilon_2 |A|^2$).

From the Clausius-Mossotti equation for an isotropic medium

$$\frac{n^2 - 1}{n^2 + 2} = \frac{4}{3} \pi \rho \alpha \quad (6)$$

where α is the polarizability, we can express the variation Δn (at the lowest order) induced by an electric field $\vec{E}$ as

$$\begin{aligned} \Delta n &= (\Delta n)_\rho + (\Delta n)_\alpha \\ (\Delta n)_\rho &= \frac{1}{2} K_\rho \lambda |A|^2 \\ (\Delta n)_\alpha &= \frac{1}{2} K_\alpha \lambda |A|^2 \\ \lambda &= \frac{2\pi}{n} \frac{c}{\omega} \end{aligned} \quad (7)$$

K_ρ, K_α are the electrostrictive and acKerr constants respectively. $(\Delta n)_\rho$ is the variation of the refractive index due to electrostriction; $(\Delta n)_\alpha$ is the variation of the refractive index arising from acKerr contribution. It should be noticed⁽⁹⁾ that only in the case of non-polar molecules it is possible to approximate the acKerr constant K_α with the contribution coming from the induced dipole moments $(K_1)_{dc}$ to the dcKerr constant; this contribution is always positive whereas the dcKerr constant K_{dc} can also be negative. For polar molecules with large dipole moments, K_α is in general smaller than the absolute value of the dcKerr constant. Again we find as it appeared from eq. 2, that the propagation of an electric wave in a nonlinear dielectric is governed by an electrostriction term and by a molecular orientational one (Kerr term). From eqs. (2) and (7) the Kerr contribution appears to depend on ω and can be strongly enhanced when ω approaches a resonant frequency of the medium.

As far as the propagation of the beam is concerned, it can be shown⁽⁷⁾ that eq. (7) implies the "self-focusing" of the beam, i.e. the beam shrinks and, after travelling in the medium a critical length

$$R_{nl} = \frac{a}{2} \left(\frac{n_0}{n_2 |A|^2} \right)^{\frac{1}{2}}$$

(a being the radius of the cross-section of the beam), gets confined in a thin waveguide ("self-trapping") provided that its power be greater than a critical value

$$P_{cr} = (1.22)^2 \lambda^2 c / 256 n_2$$

The above discussion compels us to conclude that coherent electric waves form in the medium a network of waveguides which in turn may attract "selectively" molecules in solution which have appropriate vibrational frequencies. Self-focusing strengthens very much the role of the gradient force by shrinking the beam and steepening its profile. Therefore molecules will experience stronger forces when the conditions

for the beam self-trapping are met. An increase of concentration and an alignment of the solute molecules are then generated around the beam, resulting in a dynamic self-association, which is a requirement for polymerization. It has been recently reported⁽¹⁰⁾ that polymerization can be actually induced by ordering the monomers.

At this point, let us turn to the radiation pressure term in eq.(4). The direction in which this force is exerted depends upon the longitudinal or transversal character of the time dependent electric field : a longitudinal wave produces a force transversal to the direction of propagation whereas in the transversal case the force is directed along the beam axis. As a consequence, in the first case the gradient force will change its magnitude; in the second case, molecules will also be pushed along the beam axis. The interplay between gradient forces and radiation pressure gives rise to an "optical mechanics" as shown by the experiments on laser induced "levitation" performed by Ashkin⁽¹¹⁾ and on laser isotope separation⁽¹²⁾: biological systems seem to be an excellent lab for such an "optical mechanics". The radiation pressure squeezes then the molecules, selected and attracted by the gradient force, in the waveguide and transports them along it. This squeezing enhances the probability that monomers ordered by the gradient force could polymerize. The contribution of the radiation pressure term to the assembly of the monomers could account for the dynamics of polymerization: the polymer can lose monomers at the end of the self-trapped beam while acquiring them at the opposite end.

The cytoskeleton consists of three different networks of filaments built up with protein (mainly actin and tubulin) polymers. These filaments exhibit a labile character, and their formation and dynamics strictly depend on a number of factors such as cell shape, timing, environment^(13,14). Moreover, a "treadmilling" behaviour has been observed⁽¹⁵⁾: monomers are assembled at one end of the filament and disassembled at the opposite end, giving rise to a unidirectional flow of matter. These features, insofar unexplained, can thus be accounted for by the ponderomotive forces generated by coherent electric waves propagating in form of filaments in the nonlinear medium.

An evidence for the general character of the proposed model is provided by the experiments on blood clotting⁽¹⁶⁾; filaments - the so-called contractils - appear when erythrocytes interact among themselves "à la Fröhlich" and an appropriate macromolecule is present in the suspension.

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